

# Diagnosing systematic overestimation and feature dependencies in data-driven predictions of urban CO<sub>2</sub> emissions in China

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## ABSTRACT

Machine learning is increasingly used to simulate and predict complex urban phenomena, such as CO<sub>2</sub> emission patterns in cities. However, the effectiveness, biases, and uncertainty of these predictive models are not well understood. This study systematically evaluates 18 different machine learning algorithms using a rich dataset of county-level CO<sub>2</sub> emissions in China from 2001 to 2020, incorporating 41 diverse features spanning socioeconomic, environmental, and energy consumption factors. The analysis reveals a widespread issue: nearly all models, regardless of type, tend to systematically overestimate emissions in Chinese county-level cities across both space and time. By isolating the effects of feature selection and quantifying uncertainty through bootstrap resampling, the research demonstrates that aggressive dimensionality reduction often harms model performance, while retaining a high number of features is crucial for capturing the true complexity of urban emissions. Tree-based models, such as CatBoost and LightGBM, outperform others in accuracy and uncertainty quantification. These findings highlight the urgent need for the scientific and policy communities to prioritize diagnosing and correcting systematic biases in machine learning models, rather than focusing solely on improving accuracy, to ensure reliable guidance for urban carbon management and climate policy.

## 1. Introduction

Machine learning has become a common tool for modeling complex urban spatiotemporal phenomena, such as the intricate dynamics of CO<sub>2</sub> emissions (Hsu et al., 2022; Sharifi et al., 2026; Song et al., 2025; Yap et al., 2025). To capture the complex nonlinear mechanisms driving these emissions, prediction models typically rely on high-dimensional datasets encompassing socioeconomic indicators, energy consumption profiles, land cover characteristics, and climatic conditions (Long et al., 2021; Mitchell et al., 2022; J. Xu et al., 2024). Nevertheless, as the use of these data-driven tools grows, the relative effectiveness, inherent biases, and uncertainty measurement across various algorithmic frameworks remain significantly underexplored (Y. Jin and Sharifi, 2025; Seto et al., 2014), revealing three key knowledge gaps in current literature. First, the relative effectiveness of various machine learning models in handling high-dimensional urban data has rarely been systematically compared (Y. Jin and Sharifi, 2025; Makridakis et al., 2018). Second, traditional dimensionality reduction methods are commonly used to simplify models and prevent overfitting, yet these methods may overlook synergistic features, and their actual impact on prediction models

in urban environments has not been fully investigated (Greenacre et al., 2022; Guyon and Elisseeff, 2003). Third, existing studies focus primarily on point-estimation accuracy, frequently overlooking uncertainty quantification and failing to diagnose potential structural biases (Gal and Ghahramani, 2016; Kendall and Gal, 2017; Reichstein et al., 2019).

As urban climate action initiatives progress (Aboagye and Sharifi, 2024; Dacosta Aboagye and Sharifi, 2026; Sharifi et al., 2025), understanding these methodological nuances is not merely an academic concern but has direct implications for real-world decision-making. If prediction models introduce systematic biases, their deployment in urban planning and carbon governance may lead to substantial resource misallocation or unintended regulatory burdens (Mehrabi et al., 2021; O'neil, 2017). This concern is especially salient in China, where county-level emission estimates are increasingly used to apportion emission-reduction targets under the national “dual-carbon” strategy, making systematic prediction bias a direct governance risk rather than a purely academic one (Feng et al., 2026; L. Li et al., 2026; M. T. Xu et al., 2026).

There have been some research efforts to address these issues. Indeed, recent work has formalized machine learning as a central tool for predicting urban and regional greenhouse gas emissions, but also

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exposes important methodological gaps. A systematic review of 75 machine learning -based urban GHG prediction studies (2003–2023) finds that neural networks and hybrid models dominate, generally achieving high accuracy, yet with limited cross-city comparability and model selection guidance (Y. Jin and Sharifi, 2025). Across China and other regions, studies increasingly leverage rich spatial, socioeconomic, and land-use data to resolve emissions at building, block, city, and megaregion scales (Luo et al., 2023; Luo et al., 2024; Tian et al., 2024; Yu et al., 2024; G. Zhang et al., 2024; X. Zhang et al., 2021). Comparative studies consistently show machine learning and deep learning models outperform traditional statistical approaches for CO<sub>2</sub> and air-pollution time series, including daily national emissions and regional industrial emissions (Ajala et al., 2025; X. Li and Zhang, 2023; Luo et al., 2023; Sun and Sun, 2017; Ye et al., 2024). Systematic benchmarking of 14 models across major emitting regions and of six models for China's daily emissions both conclude that tree-based ensembles and recurrent networks (especially LSTM) provide superior accuracy and robustness, though at different computational costs (Ajala et al., 2025; Luo et al., 2023). City-scale studies in China similarly report extreme gradient boosting and Extra-Trees as top performers for residential and multi-city carbon prediction and driver analysis (Luo et al., 2024; Yu et al., 2024; G. Zhang et al., 2024).

Several studies explicitly integrate dimensionality reduction or feature selection. PCA combined with regularized extreme learning machines improves national CO<sub>2</sub> forecasts for China relative to higher-dimensional baselines (Sun and Sun, 2017). A multi-stage framework using singular value decomposition, clustering, and neuro-fuzzy/ANN models for G20 emissions shows sizable gains over linear regression (Mardani et al., 2020). At urban and building scales in China, wrapper-based feature selection and LASSO-type filtering enhance ensemble models and help identify key built-environment, climate-zone, and socio-economic drivers (X. Liu et al., 2022; Luo et al., 2023; Sun and Sun, 2017; Ye et al., 2024; Yu et al., 2024; G. Zhang et al., 2024).

Reviews of ANN and deep-learning forecasting in environmental domains emphasize that input uncertainty is most often addressed,

while structural, parameter, and output uncertainties are rarely quantified (Abdar et al., 2021; Cabaneros and Hughes, 2022). Probabilistic deep-learning studies in air-quality forecasting demonstrate that Bayesian neural networks and deep ensembles can provide calibrated predictive intervals, but remain under-used in emission modeling (Abdar et al., 2021; Murad et al., 2021). Interpretable machine learning (e.g., SHAP with Extra-Trees and deep cross networks) is increasingly applied in Chinese urban studies to expose nonlinear feature effects and spatial heterogeneity, yet seldom coupled with full uncertainty quantification (X. Liu et al., 2022; Luo et al., 2024; Tian et al., 2024; Yu et al., 2024; G. Zhang et al., 2024).

Overall, the literature confirms strong gains from advanced ML and hybrid architectures for urban and regional CO<sub>2</sub> prediction, including in Chinese case studies, and documents emerging use of dimensionality-reduction and feature-selection pipelines. However, systematic cross-model comparisons on shared, high-dimensional urban datasets and integrated approaches to uncertainty and structural bias remain limited, motivating a more unified experimental framework. To address this issue and the fundamental gaps mentioned above, this study systematically evaluates 18 diverse machine learning algorithms to establish a robust comparative framework and systematically diagnose the structure of their prediction errors, in the context of Chinese county-level urban CO<sub>2</sub> emission modeling.

We constructed a comprehensive historical observation panel dataset covering county-level city CO<sub>2</sub> emissions in China from 2001 to 2020, integrating 41 features across distinct categories, as shown in Fig. 1. Our analytical pipeline rigorously evaluates a wide range of models, including tree-based models, linear and regularized models, neural networks, and kernel-based methods, examining their performance across both spatial and temporal dimensions. We isolated the effects of feature dependency by comparing models trained on the original feature set against those trained under different dimensionality reduction schemes, specifically Pearson correlation, Mutual Information (MI), and Random Forest (RF) importance (Breiman, 2001; Guyon and Elisseeff, 2003). Furthermore, we employed a non-parametric bootstrap

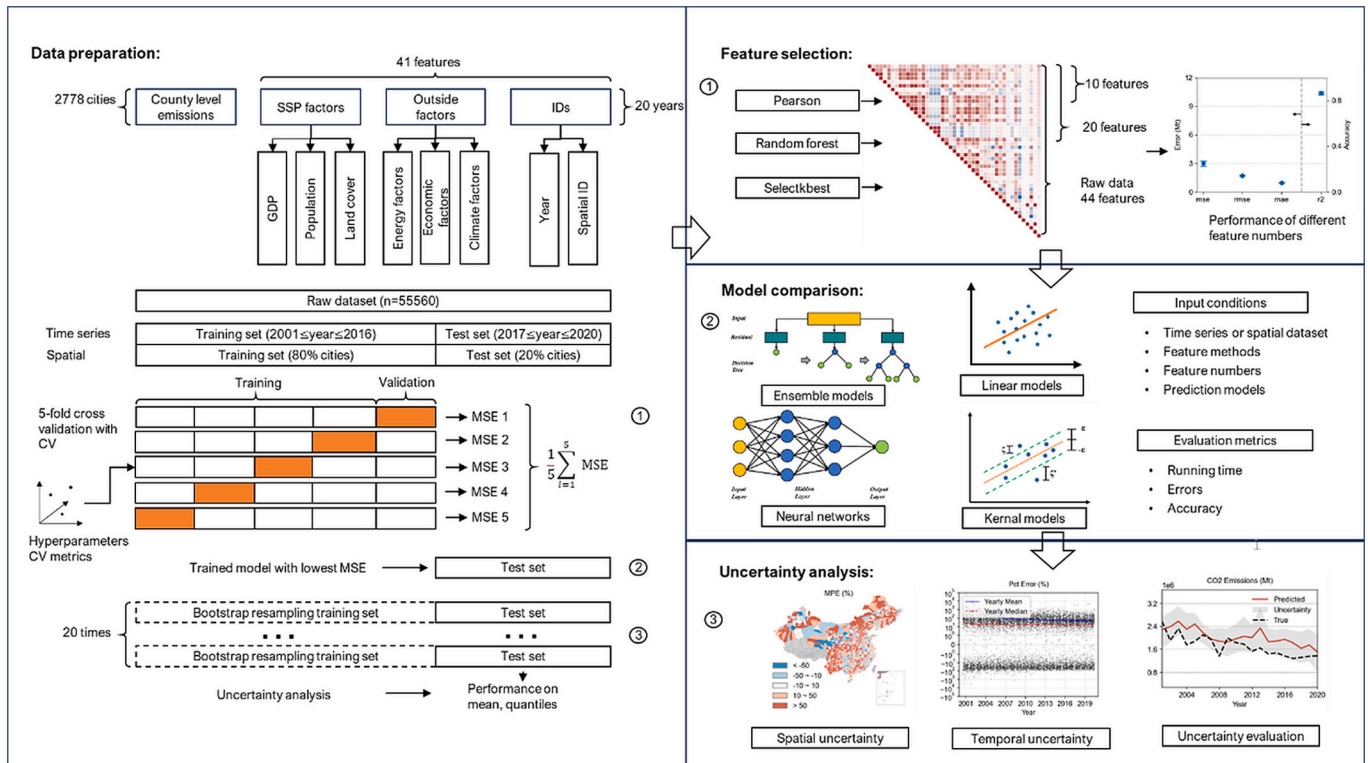


Fig. 1. Overview of different methods used in this research.

resampling approach to rigorously quantify prediction uncertainty, using Prediction Interval Coverage Probability (PICP) and Mean Prediction Interval Width (MPIW) as core evaluation metrics (Efron and Tibshirani, 1994; Khosravi et al., 2011).

Our research challenges the traditional dependence on aggressive dimensionality reduction in environmental modeling. We demonstrate that such approaches often impair model performance, while maintaining high-dimensional city features is crucial for accurately capturing complex system dynamics. Although tree-based models, especially CatBoost and LightGBM, show strengths in both point-estimation accuracy and uncertainty quantification, our analysis uncovers a common pattern: nearly all assessed models have a systematic tendency to overestimate the target variable. These findings highlight an urgent need for the scientific community to shift its focus from solely seeking higher prediction accuracy to actively diagnosing and mathematically correcting systematic model biases before these algorithms impact real-world climate interventions (Reichstein et al., 2019; Rolnick et al., 2022).

## 2. Methods

### 2.1. County-level city CO<sub>2</sub> emission estimation

The core target variable of this study is the annual CO<sub>2</sub> emissions of Chinese counties from 2001 to 2020. Due to the lack of direct, bottom-up county-level city emission inventories, we employed a spatial down-scaling technique to allocate provincial emission data from the China Carbon Accounting Databases (CEADs) to the county scale (X. Chen and Nordhaus, 2011; Shan et al., 2018). Specifically, based on county-level administrative boundaries provided by the National Geomatics Center of China, we extracted yearly 500-m resolution Nighttime Light (NTL) grid data by Chen et al. to calculate the total NTL intensity for each city (X. X. Chen et al., 2024). Provincial CO<sub>2</sub> emissions were then down-scaled according to the proportion of each city's NTL intensity relative to the provincial total (J. Zhao et al., 2019), which is built upon the scientific assumption that NTL intensity serves as a spatial proxy for anthropogenic CO<sub>2</sub> emission activities (Ghosh et al., 2010; Oda and Maksyutov, 2011).

### 2.2. Spatiotemporal prediction feature set construction

To comprehensively characterize the complex nonlinear mechanisms driving urban CO<sub>2</sub> emission dynamics, we systematically integrated 41 features (Table 1) covering socioeconomic, environmental, and energy consumption dimensions to construct a historical observation panel dataset from 2001 to 2020. All features incorporated into the models are organized into the following four categories:

**Population and macroeconomic indicators.** Economic and population bases at the urban scale are fundamental variables of CO<sub>2</sub> emissions. Gridded GDP data (30 arc-second resolution) were sourced from the global annual dataset by Kummur et al., while population data were obtained from the Gridded Population of the World (GPW) dataset at the same 30 arc-second resolution (Doxsey-Whitfield et al., 2015; Kummur et al., 2025). Given that GPW only provides population estimates at five-year intervals, we generated a continuous annual population time series using linear interpolation (Hsu et al., 2022). Additionally, we extracted provincial socioeconomic and industrial structure characteristics from Chinese statistical yearbooks, including the value-added of the three industrial sectors, import and export trade volumes, per capita income for urban and rural residents, and transport vitality indicators such as passenger volume and passenger vehicle ownership.

**Land use and air quality characteristics.** Urban morphology and environmental carrying capacity features were obtained through remote sensing and inverted grid data. Specifically, areas of cropland, forest, and urban built-up land were extracted from the MODIS 500-m resolution annual land cover product (2001–2020) (Friedl and Sulla-Menash,

**Table 1**  
Details of data preparation.

| Feature     | Detail                         | Source    | Resolution           |
|-------------|--------------------------------|-----------|----------------------|
| uid         | Unique ID                      |           |                      |
| prov_id     | Province ID                    |           |                      |
| year        | Year                           |           |                      |
| dew         | Dew point temperature          | ERA5      | grid, 0.25 degree    |
| temp        | Air temperature (2 m)          | "         | "                    |
| pop         | Population                     | NASA      | grid, 30 arc-seconds |
| pm25        | PM2.5                          | Donkelaar | grid, 0.01 degree    |
| gdp         | GDP                            | Kummur    | grid, 30 arc-seconds |
| area_crop   | Cropland area                  | Modis     | grid, 500 m          |
| area_forest | Forest area                    | "         | "                    |
| area_urban  | Urban area                     | "         | "                    |
| val_ind     | Industrial value added         | Yearbook  | Provincial data      |
| export      | Exports                        | "         | "                    |
| import      | Imports                        | "         | "                    |
| val_pri     | Primary industry value added   | "         | "                    |
|             | Secondary industry value added | "         | "                    |
| val_ter     | Tertiary industry value added  | "         | "                    |
| inc_urban   | Urban per capita income        | "         | "                    |
| inc_rural   | Rural per capita income        | "         | "                    |
| pass_vol    | Passenger traffic volume       | "         | "                    |
| cars        | Passenger car ownership        | "         | "                    |
| coal_raw    | Raw coal                       | CEADs     | "                    |
| coal_clean  | Cleaned coal                   | "         | "                    |
| briquettes  | Briquettes                     | "         | "                    |
| coke        | Coke                           | "         | "                    |
| gas_coke    | Coke oven gas                  | "         | "                    |
| oil_crude   | Crude oil                      | "         | "                    |
| gasoline    | Gasoline                       | "         | "                    |
| kerosene    | Kerosene                       | "         | "                    |
| diesel      | Diesel                         | "         | "                    |
| oil_fuel    | Fuel oil                       | "         | "                    |
| lpg         | Liquefied petroleum gas        | "         | "                    |
| gas_ref     | Refinery dry gas               | "         | "                    |
| gas_nat     | Natural gas                    | "         | "                    |
| heat        | Heat                           | "         | "                    |
| elec        | Electricity                    | "         | "                    |
| evap        | Evaporation                    | ERA5      | grid, 0.25 degree    |
| wind        | Wind speed (10 m)              | "         | "                    |
| press       | Surface Pressure               | "         | "                    |
|             | Surface Solar Radiation        | "         | "                    |
| solar       | Downwards                      | "         | "                    |
| precip      | Total Precipitation            | "         | "                    |

2022). Ambient air quality was characterized using gridded PM<sub>2.5</sub> concentration data with a spatial resolution of 0.01 degrees, developed by van Donkelaar et al. (Van Donkelaar et al., 2021).

**Meteorological dynamics.** To control the indirect effects of climatic conditions on energy systems and CO<sub>2</sub> emissions, we introduced the ERA5-Land reanalysis dataset from the European Centre for Medium-Range Weather Forecasts (ECMWF) (Hersbach et al., 2020). We extracted and calculated monthly mean meteorological features, specifically: 2-m air temperature, dewpoint temperature, surface pressure, 10-m wind speed, total precipitation, evaporation, and surface solar radiation downwards.

**Energy consumption structure.** To explicitly parameterize direct emission sources within the models (Eggleston et al., 2006), we incorporated detailed provincial total energy consumption data from the CEADs database (Z. Liu et al., 2015). This feature subset provides comprehensive coverage of 15 major fossil fuels and secondary energy types, including: raw coal, cleaned coal, other washed coal, coke, coke oven gas, crude oil, gasoline, kerosene, diesel oil, fuel oil, liquefied petroleum gas, refinery gas, natural gas, as well as heat and electricity consumption.

### 2.3. Experimental design and data preprocessing

To evaluate the performance and uncertainty quantification of various machine learning algorithms, we developed a regression framework (Fig. 1). The target variable for prediction is CO<sub>2</sub> emissions of county level cities, with 41 multidimensional features used as input features. Prior to model training, all numerical features were standardized with zero mean and unit variance (Hastie, 2009). To prevent data leakage (Kuhn and Johnson, 2013), we implemented two independent data splitting strategies for the integrated dataset. For the spatial dataset, all county-level cities were randomly assigned to an 80% training set and a 20% independent test set. For the temporal dataset, a chronological split was applied (Hyndman and Athanasopoulos, 2018), data from the 16-year period between 2001 and 2016 served as the training set, while data from 2017 to 2020 was used as the test set.

### 2.4. Feature selection protocol

To systematically evaluate model sensitivity to dimensionality reduction and feature dependency, we established a controlled comparative protocol. This step was performed only on the training set. Alongside a baseline utilizing the full high-dimensional feature set, we implemented three mathematically distinct filtering and ranking algorithms, restricting the selected subset to  $k$  features, including  $k = 10$ , and  $k = 20$ , to isolate the impact of the selection method:

**Pearson Correlation:** A univariate linear filter that ranks features based on their absolute linear correlation with the target variable (Hastie, 2009):

$$r_{x_j, y} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

**Mutual Information:** A non-parametric, entropy-based filter capable of capturing complex, non-linear dependencies by quantifying the reduction in uncertainty for the target variable given a feature (Shannon, 1948):

$$I(X_j; Y) = \int_Y \int_{x_j} p(x_j, y) \log \left( \frac{p(x_j, y)}{p(x_j)p(y)} \right) dx_j dy$$

**Random Forest Importance:** A Random Forest Regressor was fitted to the training set, and feature importance was calculated using the Mean Decrease in Impurity (MDI) (Breiman, 2001). The impurity  $i(m)$  at a given node  $m$  is measured by the variance of the target values. The importance of a feature  $j$  is the weighted sum of impurity decreases  $\Delta i$  across all nodes  $m \in M_{t,j}$  where feature  $j$  is used to split the data in tree  $t$ , averaged over the entire forest of  $N_{\text{trees}}$ :

$$FI_j = \frac{1}{N_{\text{trees}}} \sum_{t=1}^{N_{\text{trees}}} \sum_{m \in M_{t,j}} (w_m i(m) - w_{\text{left}} i(m_{\text{left}}) - w_{\text{right}} i(m_{\text{right}}))$$

where  $w_m$  is the proportion of samples reaching node  $m$ . This method is particularly advantageous as it evaluates variables not only in isolation but also accounts for complex, higher-order interactions between multiple features.

### 2.5. Model training and hyperparameter optimization

We evaluated a diverse array of machine learning paradigms to establish a robust comparative framework. The computational pipeline encompasses 18 algorithms, broadly categorized into: Tree-based models, Linear Models, Neural Networks and Kernel Models. Model descriptions are provided in Table S1 (supplementary material). To ensure an unbiased evaluation and optimal model configuration,

hyperparameter tuning was conducted using an exhaustive grid search integrated with 5-fold cross-validation (GridSearchCV). Hyperparameter details are shown in Table S2. The data was shuffled and split. The objective function for optimization across all models was the minimization of the negative Mean Squared Error (MSE).

### 2.6. Performance evaluation and uncertainty quantification

Model performance on the independent test sets was quantified using standard point-estimation metrics  $R^2$ . To rigorously assess the reliability of the predictions, we integrated a non-parametric bootstrap resampling approach for uncertainty quantification. For each optimized model,  $N$  ( $N = 20$ ) independent training sets of the same size as the original training data were generated via random sampling with replacement. The model was subsequently retrained on each bootstrap sample and deployed to predict the test set (Efron, 1992). This procedure produced a prediction distribution for each observation, from which the ensemble mean and the 95% prediction intervals (defined by the 2.5th and 97.5th percentiles) were extracted. Uncertainty quantification performance was evaluated using the PICP to assess reliability and the MPIW to evaluate precision (Khosravi et al., 2011). Systematic biases were further investigated by analyzing the spatial and temporal distribution of MPE and PE to quantify structural overestimation or underestimation tendencies across algorithms.

#### 2.6.1. PICP

This metric evaluates the proportion of true observations ( $y_{\text{true},i}$ ) that successfully fall within the prediction intervals constructed by the model. It is calculated as the average of the indicator function  $\mathbb{I}(\cdot)$  across all  $n$  samples. A higher PICP signifies that the model possesses greater reliability in capturing the true fluctuations of CO<sub>2</sub> emissions:

$$PICP = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(y_{\text{low},i} \leq y_{\text{true},i} \leq y_{\text{high},i})$$

#### 2.6.2. MPIW

While high coverage is essential, excessively wide prediction intervals provide little practical value for decision-making. Therefore, we evaluated the sharpness (or precision) of the intervals by computing the average difference between the upper and lower prediction bounds:

$$MPIW = \frac{1}{n} \sum_{i=1}^n (y_{\text{high},i} - y_{\text{low},i})$$

## 3. Results

### 3.1. Divergence in feature selection mechanisms

Our feature selection experiments reveal that, when dealing with high-dimensional city data, conventional dimensionality reduction strategies often weaken rather than enhance model generalization. In both time-series and spatial datasets (Fig. S1), univariate filtering methods such as Pearson correlation and MI led to a systematic decline in prediction accuracy. This penalty effect was even more severe in spatial prediction tasks characterized by higher heterogeneity. When the feature dimension was compressed to 10 or 20, Pearson and MI methods caused a substantial decline in the average  $R^2$  across models, ranging from 24.7% to 33.1%.

In contrast, RF-based feature selection demonstrated relative robustness. It successfully improved the performance of certain tree-based models, such as CatBoost achieving an  $R^2$  of 0.86, where 8 of the top 10 core features overlapped with RF's selections (Fig. S 2–S3). However, this advantage revealed an algorithmic homology bias (Molnar, 2020; Strobl et al., 2007). For models with different underlying architectures, such as Multi-Layer Perceptrons (MLP) or classical linear regression, the feature subsets extracted by RF failed to generalize across

models, and still resulted in a decline in the average performance of the overall model library.

We further delve into the underlying disagreements among different feature selection mechanisms (Fig. S3). Specifically, Pearson and MI focus on quantifying independent associations between individual features and CO<sub>2</sub> emissions. In contrast, ensemble algorithms like RF focus on the global nonlinear contributions of multi-dimensional features within datasets (Dai et al., 2024; Yuan et al., 2024). The data indicates that univariate filtering easily misjudges and strips away key features possessing strong nonlinear and interactive effects. For instance, “cropland area” is treated by Pearson and MI as redundant noise with ranking below 36th. However, RF identifies it as a Top 4 core feature for both temporal and spatial prediction.

Crucially, despite the severe divergence among algorithms when evaluating interactive features, they reached a consensus on the defining key variables for CO<sub>2</sub> emissions. Whether in terms of temporal evolution or spatial distribution, “urban built-up area” and “Gross Domestic Product” were consistently ranked as the top three key features by all evaluation mechanisms. Consistent with other studies, this result suggests that macroeconomic growth and urban land expansion are among the primary variables of CO<sub>2</sub> emissions (Dong et al., 2025; Zhou et al., 2025).

### 3.2. Model comparison and uncertainty quantification

As shown in Fig. 2, we evaluated the prediction performance and uncertainty quantification capabilities of 18 baseline models using original features. The results indicate that tree-based model architectures possess structural advantages when processing high-dimensional features. In time-series tasks, CatBoost achieved the highest fitting accuracy ( $R^2 = 0.84$ ), followed by LightGBM ( $R^2 = 0.82$ ) (T. Chen and Guestrin, 2016; Prokhorenkova et al., 2018). They also outperform neural networks and kernel-based models in terms of runtime (Fig. S4). In the spatial dataset, despite increased spatial heterogeneity, these tree-based models maintained robust feature extraction and generalization capabilities ( $R^2 = 0.67$ ). This suggests that tree-based model architecture can more effectively capture the complex nonlinear relationships inherent in high-dimensional city data.

In scientific forecasting, models must not only provide accurate point estimates but also reasonably quantify prediction uncertainty. An ideal prediction interval must strike a balance between reliability, reflected by high PICP, and information precision, reflected by narrow MPIW (Khosravi et al., 2011). Evaluation results show that CatBoost and LightGBM achieve the best trade-off. CatBoost provides the most conservative and reliable interval coverage across both temporal and spatial dimensions, with coverage reaching 80.7% and 63.9%, respectively. LightGBM, in contrast, produces narrower prediction intervals while

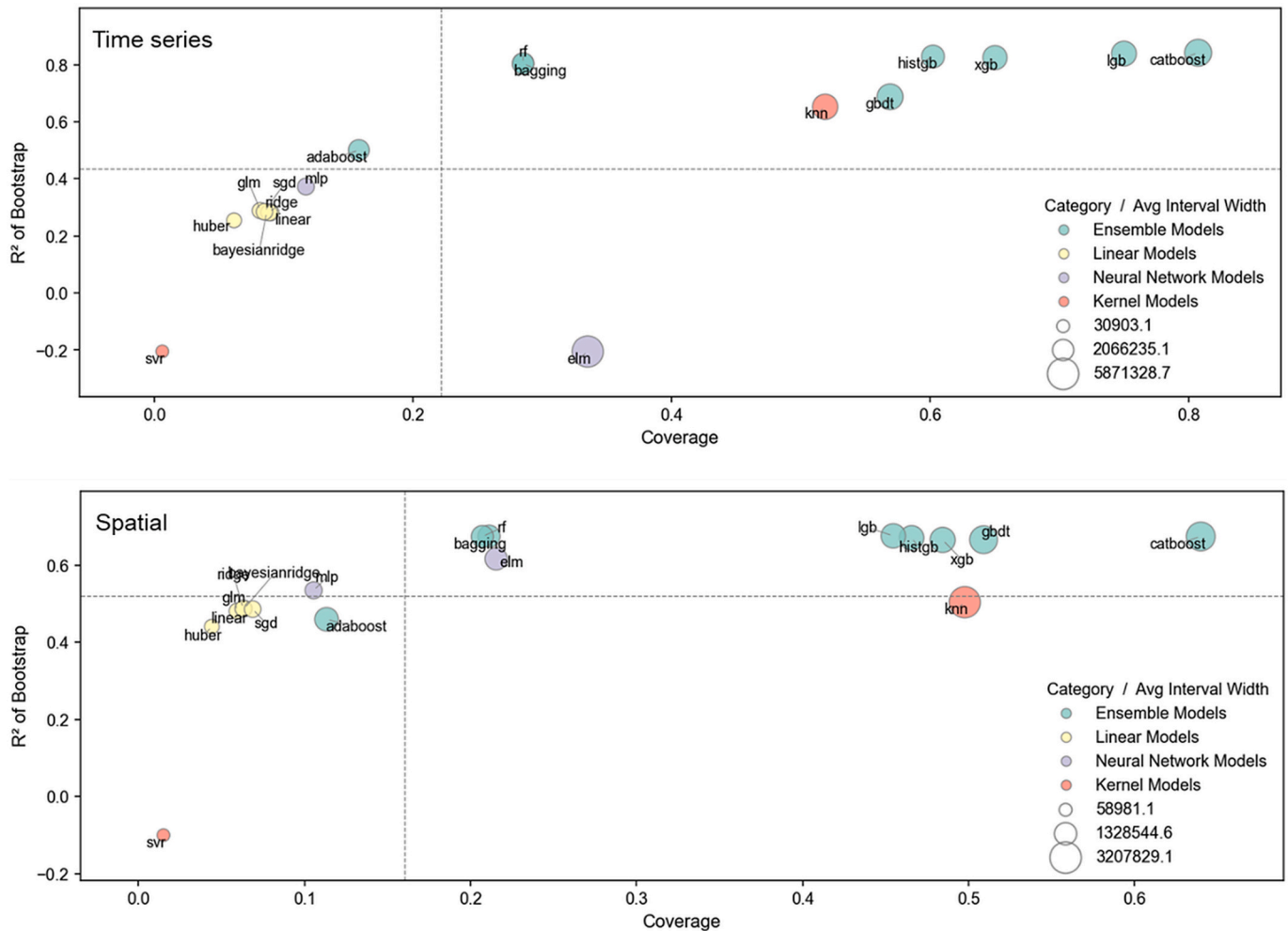


Fig. 2. Comparison of prediction accuracy and uncertainty across models.

Top and bottom panels show results for the time-series and spatial datasets, respectively. In each subplot, individual scatters represent the results of 20 bootstrap iterations for each model using the original prediction features. The y-axis represents the mean  $R^2$ , and the x-axis represents the PICP. Point size indicates the MPIW. Point colors indicate model categories, while the horizontal and vertical whiskers represent the median values for  $R^2$  and PICP, respectively.

maintaining high coverage of 75.0% for time-series data. Notably, although Bagging and RF show acceptable point-estimation accuracy, their prediction intervals are overly narrow, resulting in a substantial proportion of true values falling outside the intervals, with coverage dropping to approximately 21%–28%. This overconfidence suggests that these algorithms systematically underestimate the variability of real-world systems.

Compared to the robustness of tree-based models, traditional linear models, neural networks, and kernel methods exhibited structural limitations across all metrics (Fig. S5). Linear models, such as Ridge, are constrained by linear geometric assumptions, which causes them to lag in prediction accuracy (Hastie, 2009). They also lose their capability for uncertainty quantification. The MPIW they generated was extremely narrow, failing to account for more than 90% of the true observed variance (PICP <10%). Furthermore, non-ensemble architectures demonstrated instability across scales: the Extreme Learning Machine (ELM) showed some fitting capability on the spatial dataset ( $R^2 = 0.62$ ), but failed completely in time-series tasks. Support Vector Regression (SVR) yielded negative  $R^2$  values and extremely low PICP in both datasets. These results confirm that traditional architecture struggles to adapt to urban CO<sub>2</sub> emission predictions characterized by complex heterogeneity (Y. Jin and Sharifi, 2025; Wang and Biljecki, 2022).

### 3.3. Systematic deviations in CO<sub>2</sub> emission trajectories

As shown in Fig. 3, for the time-series dataset, we calculated the city-wise average Mean Percentage Error (MPE) of 18 models using original features from 2017 to 2020. We analyzed the mean MPE for each model, alongside the count of cities where MPE was greater or less than zero. First, most models exhibited a systematic tendency toward overestimation. Of the 18 models analyzed, 17 displayed a positive mean MPE, with the count of cities exhibiting positive MPE outnumbering those with negative MPE. AdaBoost showed the most severe overestimation, reaching a maximum mean MPE of 251.46% and producing positive errors in 2135 cities. Models such as Bagging, Random Forest (RF), and MLP also displayed strong positive prediction biases. While gradient boosting models, such as LightGBM, CatBoost, HistGB, and

XGBoost recorded relatively lower mean MPEs (ranging from 28.82% to 53.96%), they still maintained a systematic positive deviation rather than eliminating the bias. Diverging from the general trend, the Extreme Learning Machine (ELM) was the only model with a negative mean MPE (−231.81%), indicating a substantial negative bias. Regarding city distribution, Support Vector Regression (SVR) and ELM were the only two models in which the number of cities with negative MPE (1859 and 1798, respectively) far outnumbered those with positive MPE. Given their low prediction accuracy ( $R^2 < 0$ ), these results are considered unreliable.

For the spatial dataset (Fig. 4), we calculated the annual percentage error from 2001 to 2020 for 18 models across 20% of the cities (556 cities). We recorded the mean and median percentage errors for all years. First, most models showed a persistent overestimation trend over time. Thirteen out of 18 models had an annual percentage error greater than zero for all 20 years. This suggests that when processing this spatial dataset, most algorithms, especially tree-based models, and neural networks, tend to produce positive prediction bias. For instance, although tree-based models (CatBoost, LightGBM, XGBoost, and HistGB) maintained lower median percentage errors (15.28% to 19.13%), their substantially higher mean errors (approximately 216% to 309%) highlight the presence of extreme positive outliers. In contrast, linear models, such as Linear Regression and its derivatives, were uniquely distorted by extreme underestimation. Although their median errors remained positive (40% to 45%) for the majority of the period, extreme underestimations concentrated in the 2001–2005 period resulted in massive negative mean errors, with SGD reaching a mean of −1660.90%. This sharp discrepancy between mean and median values illustrates how linear models are highly vulnerable to extreme negative biases when processing early spatial data.

## 4. Discussion

While machine learning has become a standard tool for simulating complex urban spatiotemporal phenomena (Rolnick et al., 2022; L. Zhao and He, 2025), the relative effectiveness, structural biases, and uncertainty quantification of different algorithms remain under-researched

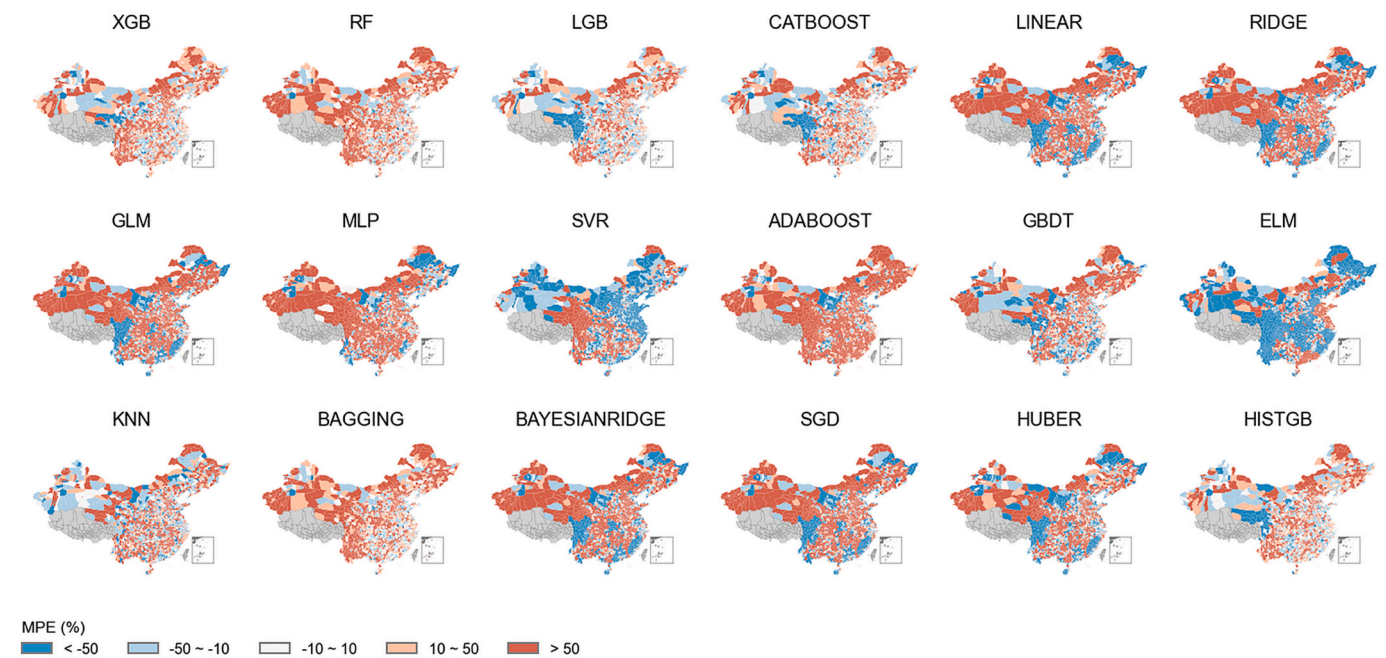
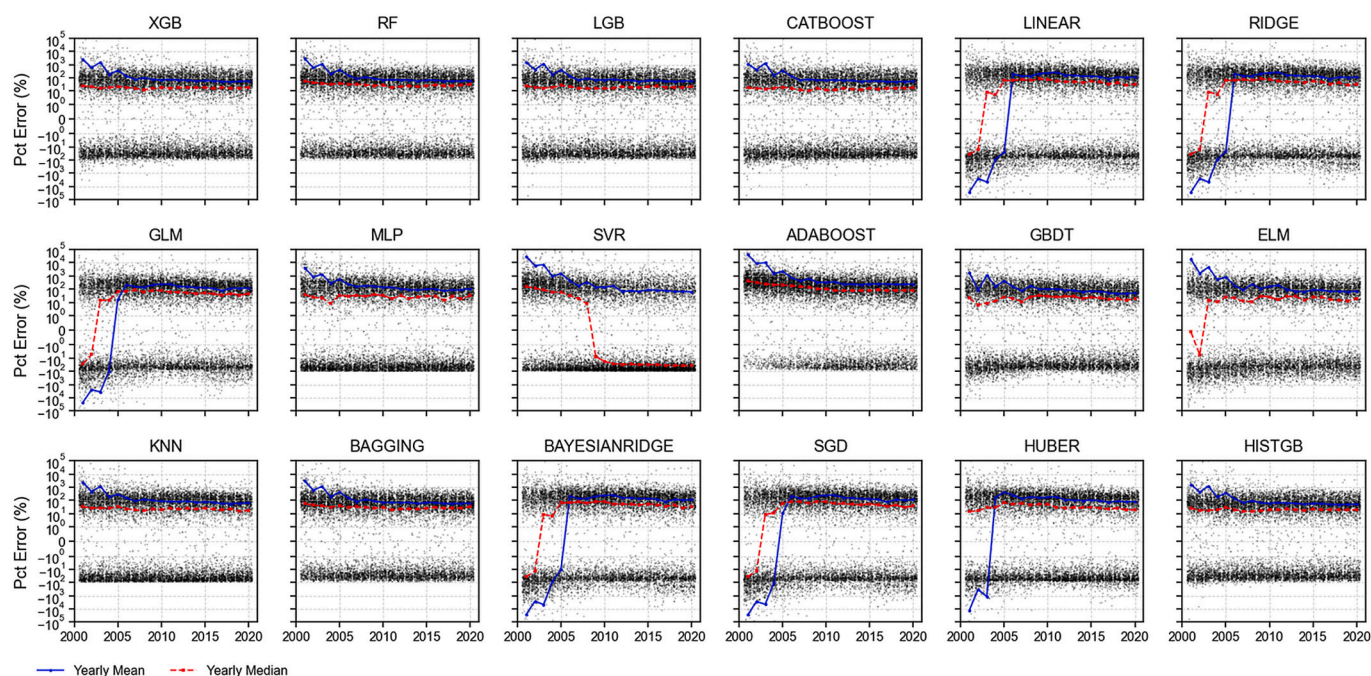


Fig. 3. Spatial uncertainty distribution of prediction models.

Each subplot represents a prediction model. We calculated the MPE for each county-level city based on the mean of 20 bootstrap predictions for the time-series test set (2017–2020). Red indicates positive MPE (overestimation relative to the ground truth), while blue indicates negative MPE (underestimation relative to the ground truth).



**Fig. 4.** Temporal uncertainty distribution of prediction models.

Each subplot represents a prediction model. We calculated the annual Percentage Error (PE) from 2001 to 2020 for 20% of the county-level cities (556 cities) in the spatial test set, based on the mean of 20 bootstrap predictions. Blue curves represent the annual mean PE, and red curves represent the annual median PE.

(Y. K. Jin et al., 2024; Pessach and Shmueli, 2023; L. Xu et al., 2021). This study systematically evaluates 18 CO<sub>2</sub> emission prediction models to establish a robust comparative framework and, more importantly, to diagnose the structure of their prediction errors across space, time, and model architecture. We find that tree-based models, such as CatBoost and LightGBM, demonstrate advantages in both point estimation accuracy ( $R^2$  up to 0.841) and uncertainty quantification. The comparatively weaker performance of the six linear and regularized baselines relative to tree-based ensembles is consistent with well-established patterns in literatures (Y. K. Jin et al., 2024; Mutz et al., 2024). In particular, we pinpoint a common systematic shortcoming: almost all the models assessed consistently show a pattern of overestimating emissions in Chinese county-level cities in both spatial and temporal contexts.

Our work builds upon extensive literature on machine learning applications in environmental and urban studies (Y. Jin and Sharifi, 2025; Zhang et al., 2023a; Y. Zhang et al., 2023b), but departs from traditional dimensionality reduction methods. We show that selecting features too aggressively often leads to a decline in model performance. Univariate filtering discards features that may be weak individually but possess strong synergistic prediction ability. Furthermore, while wrapper-based feature selection methods like RF are widely adopted, we find their accuracy gains are strictly limited to homogeneous tree-based models. In these cases, RF selection acts merely as a structural prior rather than a universal optimizer, evidenced by the high overlap of top features between RF and CatBoost, and fails entirely to enhance the performance of neural networks or classical regression models.

Given the complexity of high-dimensional city data (X. Y. Chen and Sun, 2022; Shao et al., 2023), we rigorously tested model stability through bias analysis. In the time-series analysis, 17 out of 18 models exhibited a positive MPE. Taking the top-performing LightGBM as an example, 1674 out of 2778 cities (60%) were overestimated. This positive bias persisted across the 20-year span of the spatial dataset, with overestimation in at least 15 of 20 years for all models. To ensure this was not a defect of our primary data, we validated our findings using the EDGAR CO<sub>2</sub> emission dataset (Monica Crippa et al., 2022) (Fig. S6). EDGAR is a bottom-up emissions inventory compiled using activity data and emission factors in accordance with IPCC guidelines. Based on this

independently constructed target variable, all models except for the SVR model, consistently exhibited the same systematic overestimation. This result indicates that the observed phenomenon is not specific to a particular target-construction method and is consistently validated across both datasets.

To disambiguate the potential sources of this systematic overestimation, our study design allows us to directly weigh three candidate explanations rather than leaving the question open. First, regarding target variable construction, validation against EDGAR, an inventory built independently of our NTL-based downscaling, reproduced the same overestimation pattern (Fig. S6), indicates that the bias is unrelated to the method of constructing the target variable. Second, among the 18 models, spanning linear, kernel, tree-ensemble, and neural network architectures, 17 exhibited overestimation except for ELM, showed a negative bias, but this bias is underpinned by unstable predictive performance (Fig. 2), arguing against model architecture as the primary explanation for the observed bias. Third, regarding the training data and target distribution, we examined representative low-, medium-, and high-emission counties (Fig. S5) across four best models of each type, and found that overestimation was not confined to the lower tail of the emission distribution, which is inconsistent with the bias arising from compression of low-emission counties during downscaling. Our analysis in Section 3–1 revealed that models utilizing the original set of 41 features achieved higher accuracy than those subjected to feature selection, suggesting that the quality and quantity of predictive features are potential areas for future optimization. Fourth, an ablation experiment excluding provincial-level and identifier variables (Table S3, Fig. S7) shows the median percentage error remains positive under every feature configuration for all three best-performing models, arguing against provincial-scale data leakage as the primary source of the bias.

It is important to note that our goal is to establish a comparative framework for widely used machine learning models, which necessitates defining a specific algorithmic boundary (Monier et al., 2018; Mutz et al., 2024). Consequently, this study does not evaluate parameter-intensive deep learning architectures, such as CNNs or LSTMs, nor does it include traditional statistical models like ARIMA and Grey

Models (Kontopoulou et al., 2023; Mienye et al., 2024; Taye, 2023; Tseng et al., 2001). If the underlying mechanisms of overestimation are linked to the basic architecture of standard machine learning algorithms, integrating deep learning or traditional econometric models might either exacerbate or mitigate these biases (Nazer et al., 2023; Whang et al., 2023). Understanding how these excluded models handle high-dimensional urban data remains a critical knowledge gap for future forecasting frameworks.

Our uncertainty quantification using PICP and MPIW metrics reveals differences in model reliability, yet these assessments are limited geographically and spatially (Araza et al., 2022; Wulder et al., 2022). The current results are based entirely on Chinese county-level city data. We have not explored how changes in spatial scale, such as aggregating to the provincial level or downscaling to fine-grained grid resolutions, might alter the distribution of prediction errors or interval coverage (Y. Chen et al., 2023). Furthermore, without cross-validating models on international datasets with diverse socioeconomic characteristics, we cannot definitively determine whether the observed systematic overestimation is a universal algorithmic flaw or partially driven by the specific trajectory of rapid Chinese urban expansion during the study period (Fang et al., 2022). Given the already substantial computational cost of combining GridSearchCV, 5-fold cross-validation, and bootstrap resampling across 18 models, we adopted 20 iterations as a practical trade-off; increasing this number remains a direction for future work.

Given the prediction bias even in top-performing algorithms (Mehrabi et al., 2021), our findings serve as a warning to policymakers relying on machine learning for urban planning and carbon tracking. This concern is particularly acute in China, where county-level emission predictions are increasingly used to allocate emission-reduction targets under the national dual-carbon strategy. Systematic overestimation could result in disproportionately heavy reduction mandates being imposed on counties, undermining the equity and effectiveness of local carbon governance and causing misallocation of limited abatement resources. Accordingly, counties with relatively high predicted emissions could prioritize industrial restructuring and energy-efficiency improvements, whereas counties with lower predicted emissions could focus on maintaining low-carbon development and strengthening emissions monitoring. These differentiated measures should also consider local socioeconomic conditions and prediction uncertainty to avoid imposing disproportionate mitigation burdens. If these models are applied directly to policy formulation, systematic overestimation of emissions or other urban metrics could lead to resource misallocation or excessive regulatory burdens (O'neil, 2017). Our findings highlight the pressing necessity to move beyond simply aiming for greater accuracy and instead focus on identifying and addressing biases inherent in the model's structure (Chernozhukov et al., 2018). Future research should prioritize identifying the root causes of algorithmic overestimation, testing prediction frameworks across different global scales, and developing robust methods to debias these models before they influence real-world interventions.

## 5. Conclusion

This paper systematically evaluated the biases and performance of 18 machine learning models in predicting county-level urban CO<sub>2</sub> emissions in China from 2001 to 2020. The primary objective was to compare model effectiveness, diagnose systematic biases, and assess the impact of feature selection on prediction accuracy and uncertainty quantification.

Findings that nearly all models, especially tree-based algorithms like CatBoost and LightGBM, systematically overestimate emissions in Chinese county-level cities across both spatial and temporal dimensions. Aggressive dimensionality reduction through feature selection was shown to degrade model performance, while retaining high-dimensional features was critical for capturing complex urban emission dynamics. Tree-based models outperformed others in both point-estimation

accuracy and uncertainty quantification, but their positive prediction bias persisted.

These results have important policy implications: if machine learning models with systematic overestimation biases are used directly in urban planning or carbon governance, they could lead to resource misallocation or excessive regulatory burdens. Policymakers should therefore prioritize the diagnosis and correction of systematic model biases over the pursuit of higher prediction accuracy alone.

The study is limited by its focus on standard machine learning algorithms and Chinese county-level data, excluding deep learning and traditional statistical models, as well as international datasets. Future research should investigate the root causes of algorithmic overestimation, test models across different spatial scales and global contexts, and develop robust debiasing methods to ensure reliable application in real-world climate interventions.

## Code availability

The code of the proposed model is publicly available at GitHub: <https://github.com/UrbanModeling/model-comparison>

## Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Claude to check the grammar, punctuation, and clarity. After using this tool, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eiar.2026.108770>.

## Data availability

The raw data supporting the findings of this study are derived from publicly available databases and published resources. Provincial CO<sub>2</sub> emission inventories and specific energy consumption data were obtained from the China Carbon Accounting Databases (CEADs) (Shan et al., 2018). The EDGAR CO<sub>2</sub> emissions grid data was from the EDGAR team (M Crippa et al., 2020). Administrative boundary data are available from the National Geomatics Center of China (NGCC). GDP and population features were sourced from the global gridded GDP dataset by Kummu et al. (2025) and the Gridded Population of the World (GPW) dataset (Doxsey-Whitfield et al., 2015), respectively. Environmental and meteorological variables were extracted from the MODIS annual land cover product (Friedl and Sulla-Menashe, 2022), the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 land reanalysis dataset (Hersbach et al., 2020), and the gridded PM2.5 dataset developed by Van Donkelaar et al. (2021). Standard socioeconomic indicators are available in the respective Chinese provincial statistical yearbooks.

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